

C-Reactive Protein (CRP), High Sensitivity (Cardiac Risk Assessment)

Test Details

METHODOLOGY

Immunochemiluminometric assay (ICMA)

RESULT TURNAROUND TIME

Within 1 day

Turnaround time is defined as the usual number of days from the date of pickup of a specimen for testing to when the result is released to the ordering provider. In some cases, additional time should be allowed for additional confirmatory or additional reflex tests. Testing schedules may vary.

Use

Measurement of CRP by high sensitivity CRP assays may add to the predictive value of other markers used to assess the risk of cardiovascular and peripheral vascular disease.²⁻⁸

Special Instructions

State patient's sex on the test request form.

Limitations

Increases in CRP values are nonspecific. CRP is an indicator for a wide range of disease processes and should not be interpreted without a complete clinical history. Recent medical events resulting in tissue injury, infections, or inflammation, which may cause elevated CRP levels, should also be considered when interpreting results. Serial analysis of CRP should not be used to monitor the effects of treatment.

Specimen Requirements

Specimen

Serum **or** plasma

Volume

1 mL

Minimum Volume

0.5 mL (**Note:** This volume does **not** allow for repeat testing.)

Container

Red-top tube, gel-barrier tube, lavender-top (EDTA) tube, **or** green-top (heparin) tube

Collection Instructions

Separate serum or plasma from cells within one hour of collection.

Stability Requirements

Temperature	Period
Room temperature	7 days
Refrigerated	14 days
Frozen	14 days
Freeze/thaw cycles	Stable x3

Reference Range

Average hs-CRP level:¹

- Low risk: <1.00 mg/L
- Average risk: 1.00–3.00 mg/L
- High risk: >3.00 mg/L

Storage Instructions

Room temperature

Causes for Rejection

Gross lipemia

Footnotes

1. Pearson TA, Mensah GA, Alexander RW, et al. Markers of inflammation and cardiovascular disease: Application to clinical and public health practice: A statement for healthcare professionals from the Centers for Disease Control and Prevention and the American Heart Association. *Circulation*. 2003; 107(3):499-511. PubMed 12551878

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3. Ridker PM, Cushman M, Stampfer MJ, et al. Inflammation, aspirin, and the risk of cardiovascular disease in apparently healthy men. *N Engl J Med*. 1997; 336(14):973-979. PubMed 9077376
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(<http://www.ncbi.nlm.nih.gov/pubmed/9727541>)
5. Ridker PM, Glynn RJ, Hennekens CH. C-Reactive protein adds to the predictive value of total and HDL cholesterol in determining risk of first myocardial infarction. *Circulation*. 1998; 97(20):2007-2011. PubMed 9610529 (<http://www.ncbi.nlm.nih.gov/pubmed/9610529>)
6. Ridker PM, Rifai N, Peffer MA, et al, for the Cholesterol and Recurrent Events (CARE) Investigators. Inflammation, pravastatin, and the risk of coronary events after myocardial infarction in patients with average cholesterol levels. *Circulation*. 1998; 98(9):839-844. PubMed 9738637
(<http://www.ncbi.nlm.nih.gov/pubmed/9738637>)
7. Tracy RP, Lemaitre RN, Psaty BM, et al. Relationship of C-reactive protein to risk of cardiovascular disease in the elderly: Results from the Cardiovascular Health Study and the Rural Health Promotion Project. *Arterioscler Thromb Vasc Biol*. 1997; 17(6):1121-1127. PubMed 9194763
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8. Macy EM, Hayes TE, Tracy RP. Variability in the measurement of C-reactive protein in healthy subjects: Implications for reference interval and epidemiological applications. *Clin Chem*. 1997; 43(1):52-58. PubMed 8990222 (<http://www.ncbi.nlm.nih.gov/pubmed/8990222>)